CHAPTER 2 · DERIVATIONS

KINEMATICS

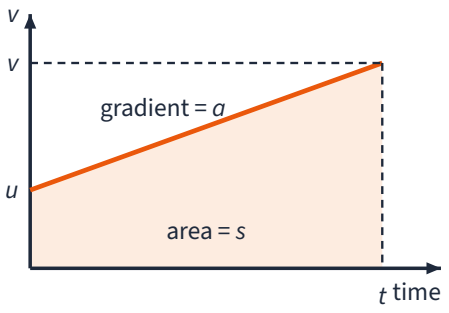
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derivation the STPM 2026 syllabus asks you to derive, deduce or show

1 EQUATIONS OF MOTION WITH CONSTANT ACCELERATION

2.1(A)

RESULT $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s = \frac{1}{2}(u + v)t$

1. Constant acceleration: $a = \frac{dv}{dt}$ is constant. Integrate from $v = u$ at $t = 0$: $\int dv = \int a\,dt$, so $\mathbf{v = u + at}$ … (1)
2. $v = \frac{ds}{dt} = u + at$. Integrate from $s = 0$ at $t = 0$: $\mathbf{s = ut + \frac{1}{2}at^2}$ … (2)
3. From (1), $t = \frac{v - u}{a}$. Substitute into (2): $s = \frac{u(v - u)}{a} + \frac{\frac{1}{2}a(v - u)^2}{a^2}$.
4. Multiply by 2a: $2as = 2uv - 2u^2 + v^2 - 2uv + u^2$, so $\mathbf{v^2 = u^2 + 2as}$ … (3)
5. Area under the v–t graph (a trapezium) gives $\mathbf{s = \frac{1}{2}(u + v)t}$ … (4)



On the v–t graph, the gradient is a and the area is s.